COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 Data Structures and Algorithms	**SECTION:**	CPE21S1
DATE PERFORMED:	7/9/2026	**DATE SUBMITTED:**	7/14/2026
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ACTIVITY NO. _1__ [REVIEW OF C++ PROGRAMMING

1. PROCEDURE OUTPUT

ILO A: Create Code That Follows the Basic C++ Code Structure

For this activity, you have to demonstrate the use of a function prototype. The section on class declaration and method definition will be used for the function prototype and the function will be defined in the follow method definition section after the main function.

A function prototype in c++ is a declaration of the name, parameters and return type of the function before its definition. Write a C++ code the satisfies the following:

- Create a function that will take two numbers and display the sum.
- Create a function that will return whether variable A is greater than variable B.
- Create a function that will take two Boolean values and display the result of all logical operations then return true if it was a success.

SECTIONS	ANSWERS
Header File Declaration Section	`#include <iostream>`

Global Declaration Section	`void sum_num(int num1, int num2);`
	`void maxx(int num1, int num2);`
	`void AND_operations(char a, char b);`
	`void OR_operations(char a, char b);`
	`void NOT_operations(char a, char b);`

Class Declaration and Method Definition Section	NONE

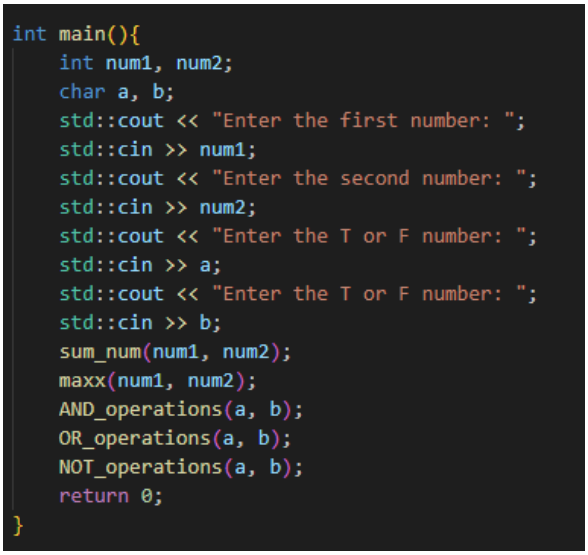
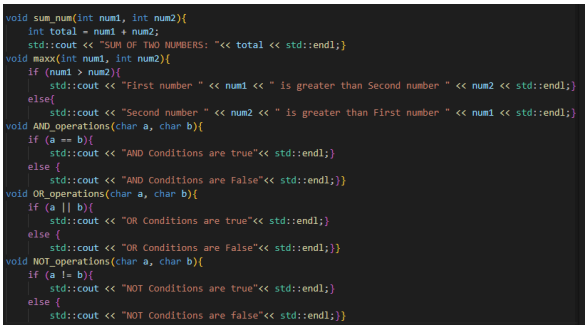
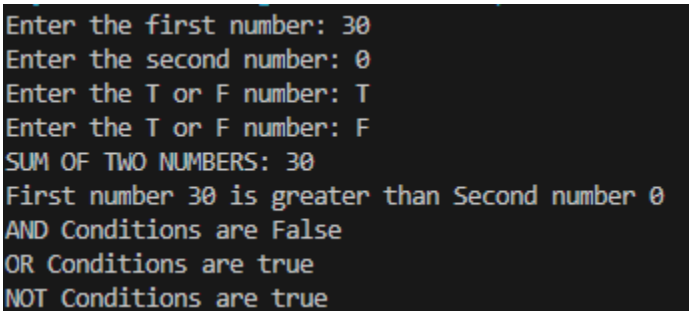
Main Function	
Method Definition	

Table 1-1. CODE SNIPPET



- Base on this output, first i ask for the user input for the number 1 and number 2 for the sum and the greater value, and then a and b for the boolean true and false. After that it shows the output for the sum of two numbers and output for the greater values and for the boolean logical operations, it show the output whether it is true or false that depends on the user input.

Table 1-1. OUTPUT

ILO B: Implement Appropriate Class Definition and Instances Based on Given Requirements

In this section, the initial implementation for a class triangle will be implemented. The step-by-step procedure is shown below:

Step 1. Include the necessary header files. For this one, we only need #include <iostream>
Step 2. Create the triangle class. Assign it with private variables: totalAngle, angleA, angleB, and angleC.

```
class Triangle{
private:
    double totalAngle, angleA, angleB, angleC;
```

Step 3. We then create public methods. The constructor must allow for creation of the object with 3 initial angles to be stored in our previously defined variables angleA, angleB and angleC. Another method has to be made if the user wants to change the initial values, this will also accept 3 arguments to change the values in angleA, angleB and angleC. Lastly, a function to validate whether the given values make our shape an actual triangle.
public:

```
public:
    Triangle(double A, double B, double C);
    void setAngles(double A, double B, double C);
    const bool validateTriangle();
};
```

Step 4. Define the methods.

```
Triangle::Triangle(double A, double B, double C) {
    angleA = A;
```



```
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}

void Triangle::setAngles(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}

const bool Triangle::validateTriangle() {
    return (totalAngle <= 180);
}
```

Step 5. Create the driver code.

```
int main(){
    //driver code
    Triangle set1(40, 30, 110);
    if(set1.validateTriangle()){
        std::cout << "The shape is a valid triangle.\n";
    } else {
        std::cout << "The shape is NOT a valid triangle.\n";
    }

    return 0;
}
```

SECTIONS	ANSWERS
Header File Declaration Section	`// Online C++ compiler to run C++ program online` `#include <iostream>`
Global Declaration Section	NONE

Class Declaration and Method Definition Section	
Main Function	
Method Definition	

```
4  class Triangle{
5      private:
6          double total_angle, angleA, angleB, angleC;
7      public:
8          Triangle(double A, double B, double C);
9          void setAngles(double A, double B, double C);
10         const bool ValidateTriangle();
```

```
12 int main() {
13     double ang1, ang2, ang3;
14     std::cout << "Enter the first side: ";
15     std::cin >> ang1;
16     std::cout << "Enter the second side: ";
17     std::cin >> ang2;
18     std::cout << "Enter the third side: ";
19     std::cin >> ang3;
20     Triangle set1(ang1,ang2,ang3);
21     if (set1.ValidateTriangle()){
22         std::cout << "The shape is a valid Triangle" << std::endl;
23     } else {
24         std::cout << "The shape is NOT a valid triangle. \n";
25     }
26     return 0;
27 }
```

```
28 Triangle::Triangle(double A, double B, double C){
29     angleA = A;
30     angleB = B;
31     angleC = C;
32     total_angle = A + B + C;
33 }
34
35 void Triangle::setAngles(double A, double B, double C){
36     angleA = A;
37     angleB = B;
38     angleC = C;
39     total_angle = A + B + C;
40 }
41
42 const bool Triangle::ValidateTriangle() {
43     return total_angle <= 180.0;
```

Table 1-2

.

```
Output

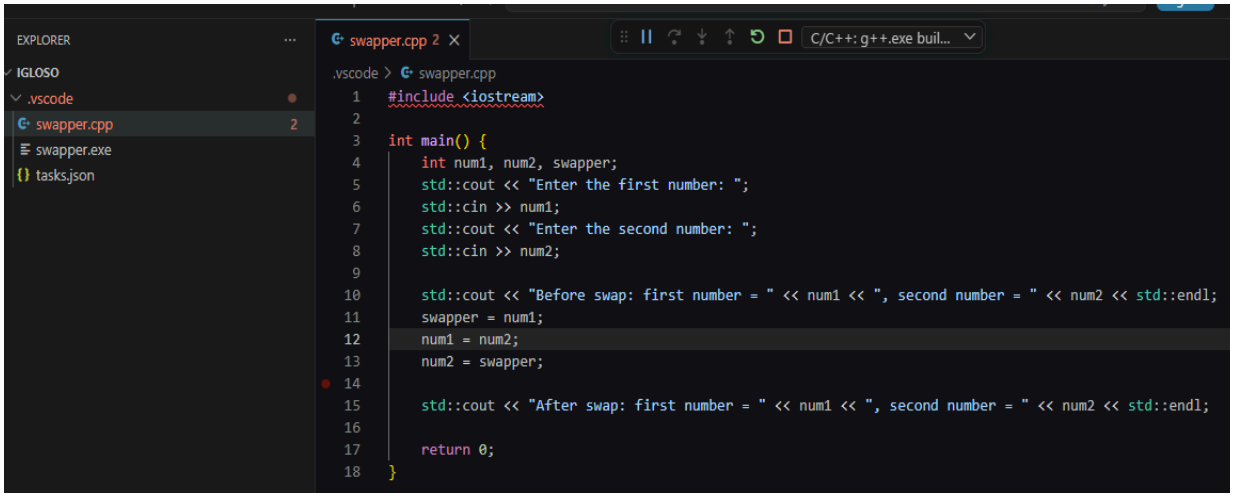
Enter the first side: 30
Enter the second side: 30
Enter the third side: 30
The shape is a valid Triangle
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

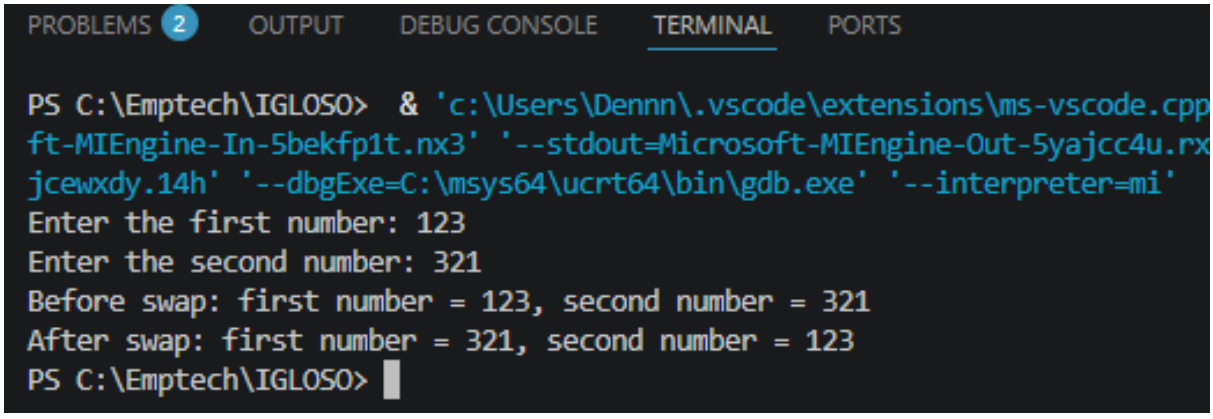
ILO C: Solve Different Problems using the C++ Programming Language

The supplementary activities are meant to gauge your ability in using C++. The problems below range from easy to intermediate to advanced problems. Note your difficulties after answering the problems below.

1. Create a C++ program to swap the two numbers in different variables.



OUTPUT



2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.

swapper.cpp | distance.cpp | Faren.cpp 2

.vscode > Faren.cpp

```cpp
#include <iostream>


int main() {
    double kelvin, a, Fahrenheit, b;
    double const1 = 1.8, const2 = 273.15, const3 = 32;
    std::cout << "Enter temperature in Kelvin: ";
    std::cin >> kelvin;
    a = kelvin - const2;
    b = const1 * a;
    Fahrenheit = b + const3;

    std::cout << "Fahrenheit = " << Fahrenheit << std::endl;

    return 0;
}
```

OUTPUT

PROBLEMS 2 | OUTPUT | DEBUG CONSOLE | TERMINAL | PO

```
PS C:\Emptech\IGLOSO> & 'c:\Users\Dennn\.vscode\ext
ft-MIEngine-In-my11uxdu.era' '--stdout=Microsoft-MIE
dsnw4hh.kzd' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe'
Enter temperature in Kelvin: 403
Fahrenheit = 265.73
PS C:\Emptech\IGLOSO>
```

3. Create a C++ program that has a function that will calculate the distance between two points

swapper.cpp 2 | distance.cpp 2

.vscode > distance.cpp

```cpp
#include <iostream>

int main(){
    double x1, x2, y1, y2, distance;
    std::cout << "Enter the x1 value: ";
    std::cin >> x1;
    std::cout << "Enter the x2 value: ";
    std::cin >> x2;
    std::cout << "Enter the y1 value: ";
    std::cin >> y1;
    std::cout << "Enter the y2 value: ";
    std::cin >> y2;

    double dx = x2 - x1;
    double dy = y2 - y1;
    double squared = (dx * dx) + (dy * dy);

    double x = squared;
    double y = 1.0;
    double e = 0.000001;
    while (x - y > e) {
        x = (x + y) / 2.0;
        y = squared / x;
    }
    distance = x;

    std::cout << "The Distance is: " << distance << std::endl;
}
```

PROBLEMS 4 | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
PS C:\Emptech\IGLOSO>  & 'c:\Users\Dennn\.vscode\exten
ft-MIEngine-In-htbefwqj.5q3' '--stdout=Microsoft-MIEng
zzogk21.m5x' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '
Enter the x1 value: 3
Enter the x2 value: 7
Enter the y1 value: 9
Enter the y2 value: 4
The Distance is: 6.40312
PS C:\Emptech\IGLOSO>
```

4. Modify the code given in ILO B and add the following functions:
   a. A function to compute for the area of a triangle
   b. A function to compute for the perimeter of a triangle
   c. A function that determines whether the triangle is acute-angled, obtuse-angled or 'others.'

swapper.cpp | distance.cpp | Faren.cpp | ILOB.cpp 2

.vscode > ILOB.cpp

```cpp
#include <iostream>
#include <cmath>

class Triangle{
    private:
        double total_angle, angleA, angleB, angleC;
    public:
        Triangle(double A, double B, double C);
        void setAngles(double A, double B, double C);
        const bool ValidateTriangle();
};

void Convertion(double A, double B, double C){
    double rad1 = A * 0.01745329251;
    double rad2 = B * 0.01745329251;
    double rad3 = C * 0.01745329251;

    double sideA = 10.0 * std::sin(rad1);
    double sideB = 10.0 * std::sin(rad2);
    double sideC = 10.0 * std::sin(rad3);

    double perimeter = sideA + sideB + sideC;
    double area = 0.5 * sideA * sideB * std::sin(rad3);

    std::cout << "Perimeter: " << perimeter << std::endl;
    std::cout << "Area: " << area << std::endl;
}

```

swapper.cpp | distance.cpp | Faren.cpp | ILOB.cpp 2

.vscode > ILOB.cpp

```cpp
void Convertion(double A, double B, double C){

int main() {
    double ang1, ang2, ang3;
    std::cout << "Enter the first angle: ";
    std::cin >> ang1;
    std::cout << "Enter the second angle: ";
    std::cin >> ang2;
    std::cout << "Enter the third angle: ";
    std::cin >> ang3;
    Triangle set1(ang1,ang2,ang3);
    if (set1.ValidateTriangle()){
        std::cout << "The shape is a valid Triangle" << std::endl;
        Convertion(ang1,ang2,ang3);
    } else {
        std::cout << "The shape is NOT a valid triangle. \n";
    }
    return 0;
}
Triangle::Triangle(double A, double B, double C){
    angleA = A;
    angleB = B;
    angleC = C;
    total_angle = A + B + C;
}

```

swapper.cpp | distance.cpp | Faren.cpp | ILOB.cpp 2

.vscode > ILOB.cpp

```
46  Triangle::Triangle(double A, double B, double C){
52
53  void Triangle::setAngles(double A, double B, double C){
54      angleA = A;
55      angleB = B;
56      angleC = C;
57      total_angle = A + B + C;
58  }
59
60  const bool Triangle::ValidateTriangle() {
61      return total_angle <= 180.0;
62  }
```

OUTPUT

PROBLEMS 2 | OUTPUT | DEBUG CONSOLE | TERMINAL | POR

```
ft-MIEngine-In-yublkd0w.vra' '--stdout=Microsoft-MIE
zcx2dpz.ycg' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe'
Enter the first angle: 45
Enter the second angle: 32
Enter the third angle: 60
The shape is a valid Triangle
Perimeter: 21.0305
Area: 16.2254
```

CONCLUSION

- Overall, I sort of managed to finish all the tasks that were given to me, and honestly I think it went pretty smoothly. I wrote C++ programs for the distance formula, temperature conversions, and also a geometric triangle class, and I believe I did well, since everything works perfectly now. In the procedure and supplementary activity, I learned how to put math into my code, like converting degrees to radians so the sine function actually works. I also learned how to untangle compilation errors, for example, fixing function declarations and the occasional wrong syntax that are useful. I used gemini only as a kind of code assistance thing, more like to double-check if my Kelvin to Fahrenheit formulas were correct, and to verify the radian calculations I used for the perimeter and area. That check helped me catch logic

errors. For my areas of improvement, I need to practice more with C++ class structures and I should pay closer attention to how functions pass variables before I just start writing my code, because I kind of a rushed a few times.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Code assistance only**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I used gemini for code assistance only, specifically on the formula of distance and how I can create it in c++ code. I also ask to check whether my formula on kelvin to fahrenheit and the conversion of radians to calculate the perimeter and area are correct. Overall I manage to complete the tasks that are given.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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